

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A Case Study

Code of Conduct:

I M. Phanith Chandu (192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Phanith Chandu

Q. No	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)				Total Marks (Out of 50)		

Annexure A

Case Study

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

1. Hough Transform for Shape Detection

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds.

Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

2. PCA in Face Recognition

A face recognition system processes high-dimensional image data, resulting in slow performance.

Analyze how PCA helps in reducing dimensionality while preserving important features.

3. Edge Detection in Low Contrast Images

An image has low contrast, causing edge detection algorithms to fail. Analyze the problem and explain how preprocessing enhances edge detection results.

4. Corner Detection in Textured Regions

A system applies corner detection in highly textured regions, resulting in excessive keypoints.

Analyze the issue and suggest methods to improve feature selection.

5. Feature Matching with Noise

Feature matching between two images fails due to noise in one image. Analyze the impact of noise and explain how preprocessing improves matching accuracy.

Case Study 1: Hough Transform for Shape Detection

1. Scenario

A system uses the Hough Transform to detect circular shapes but fails in complex backgrounds.

2. Understanding of the Case

- The Hough Transform relies on a clean edge map; complex backgrounds introduce large numbers of irrelevant edges.
- These extra edges generate false votes in the accumulator space, causing false circle detections or missed true circles.
- Overlapping textures and clutter reduce the strength of the true accumulator peak relative to noise peaks.

3. Application of Concepts

- Circle detection depends on an accurate edge/gradient map (via cv2.HoughCircles built on Canny internally).
- The Hough accumulator votes on (center_x, center_y, radius); noisy edges spread votes and weaken genuine peaks.
- Parameter tuning (param1 for the Canny threshold, param2 for the accumulator threshold, minDist, minRadius, maxRadius) directly controls sensitivity to background clutter.

4. Analysis

In a complex background, the raw edge map contains contours from textures, shadows, and unrelated objects, not just the circular target. This dilutes the accumulator votes so that the true circle's peak may not stand out enough, leading to missed detections (if param2 is too high) or many false positives (if param2 is too low). The problem is fundamentally one of signal-to-noise ratio in the edge map feeding into the transform.

Cause	Effect on Detection
Excess background edges	False circle detections (false positives)
Weak / broken object edges	True circles missed (false negatives)
Poor parameter tuning	Over- or under-sensitive detection

5. Solution Design

- Apply Gaussian blur before edge detection to suppress fine background texture and noise.
- Use adaptive or higher Canny thresholds (param1) so only strong, relevant edges are retained.
- Restrict the search range using minRadius/maxRadius based on prior knowledge of expected object size.
- Increase minDist between circle centers to avoid duplicate detections from clustered false votes.
- Apply background subtraction or ROI cropping to isolate the region of interest before running Hough Circle detection.
- Raise the accumulator threshold (param2) to require stronger evidence before accepting a detected circle.

6. Conclusion

With proper preprocessing (blurring, ROI restriction, background suppression) and careful tuning of the Canny and accumulator thresholds, the Hough Transform can be made significantly more robust to complex backgrounds, reducing both false positives and missed detections.

Case Study 2: PCA in Face Recognition

1. Scenario

A face recognition system processes high-dimensional image data, resulting in slow performance.

2. Understanding of the Case

- Each face image, when flattened, produces a very high-dimensional feature vector (e.g. a 100x100 image gives 10,000 dimensions).
- High dimensionality increases storage, computation time, and the risk of overfitting (curse of dimensionality), especially with a limited number of training faces.
- Much of this raw pixel data is redundant, since neighboring pixels and facial regions are highly correlated.

3. Application of Concepts

- PCA (used classically in face recognition as 'Eigenfaces') computes the covariance matrix of the face dataset and extracts eigenvectors (principal components) capturing the directions of maximum variance.
- Each face can then be represented as a weighted combination of a small number of top eigenfaces instead of raw pixels.
- Only the top-k components (e.g. those capturing 95% of variance) are retained, drastically reducing dimensionality.

4. Analysis

By projecting each face image onto a reduced set of principal components, PCA compresses the feature representation from thousands of pixel dimensions down to a much smaller number (often tens to a few hundred), while retaining the components that carry the most discriminative facial variation (e.g. differences in shape, shading, and structure across faces). This reduces the computational load for matching/classification and removes redundant, highly correlated pixel information that added little recognition value.

Metric	Before PCA	After PCA
Feature Dimensions	High (e.g. 10,000+ pixels)	Low (e.g. 50-150 components)
Redundancy	High (correlated pixels)	Low (uncorrelated components)
Matching Speed	Slow	Fast
Key Information	Includes noise and redundancy	Concentrated in top components

5. Solution Design

- Compute eigenfaces from the training set and select the number of components that retain ~95% of total variance.
- Project every new face image onto this reduced eigenface space before matching, instead of comparing raw pixels.
- Combine PCA with a fast distance-based classifier (e.g. nearest neighbor in the reduced space) for real-time recognition.
- Periodically re-train the PCA basis if the dataset of faces changes significantly.

6. Conclusion

PCA addresses the slow performance of the face recognition system by reducing high-dimensional pixel data to a compact set of principal components that preserve the most important facial variation, enabling faster and more efficient matching without significant loss of recognition accuracy.

Case Study 3: Edge Detection in Low Contrast Images

1. Scenario

An image has low contrast, causing edge detection algorithms to fail.

2. Understanding of the Case

- Edge detectors (Sobel, Canny, etc.) rely on detecting significant changes in intensity (gradients) between neighboring pixels.
- In a low-contrast image, the intensity difference across true object boundaries is small, so gradient magnitudes are weak.
- Weak gradients fall below typical thresholds, causing edges to be missed or detected as broken, fragmented lines.

3. Application of Concepts

- Gradient-based operators compute G_x and G_y from local pixel differences; when contrast is low, these gradients approach the noise floor and cannot be reliably separated from sensor/background noise.
- Canny's double-thresholding stage is especially sensitive here: thresholds calibrated for normal-contrast images will discard genuine but weak edges in a low-contrast image.
- Preprocessing techniques that redistribute or stretch the intensity range can amplify these weak gradients before edge detection is applied.

4. Analysis

Low contrast compresses the image's intensity histogram into a narrow range, so the difference between object and background (or between adjacent objects) becomes small relative to noise. This directly weakens the gradient signal that edge detectors depend on, leading to missed boundaries, fragmented edges, or edges only detected in the highest-contrast regions of the image.

Preprocessing Technique	Effect on Edge Detection
Histogram Equalization	Redistributes intensities, boosts overall contrast
CLAHE (adaptive equalization)	Enhances local contrast without over-amplifying noise
Gamma Correction	Expands intensity range in dark/bright regions
Contrast Stretching	Linearly stretches intensity range to full scale

5. Solution Design

- Apply histogram equalization or CLAHE to enhance local contrast before running the edge detector.
- Use gamma correction or contrast stretching to expand the intensity range, particularly in the region containing the object of interest.
- Lower or adaptively set the Canny thresholds to account for weaker gradient magnitudes after enhancement.
- Apply mild denoising (e.g. Gaussian blur) after contrast enhancement, since boosting contrast can also amplify noise.

6. Conclusion

Preprocessing the image with contrast enhancement techniques such as CLAHE or histogram equalization restores sufficient gradient strength at true object boundaries, allowing standard edge detection algorithms to reliably detect edges that would otherwise be missed in the original low-contrast image.

Case Study 4: Corner Detection in Textured Regions

1. Scenario

A system applies corner detection in highly textured regions, resulting in excessive keypoints.

2. Understanding of the Case

- Corner detectors (e.g. Harris, Shi-Tomasi) respond to any local region with significant intensity variation in multiple directions.
- Highly textured regions (foliage, fabric patterns, gravel, etc.) contain many small local variations that satisfy the corner response criteria, even though they are not structurally meaningful features.
- This produces an excessive number of keypoints, many of which are redundant, unstable, or not useful for downstream tasks like matching or tracking.

3. Application of Concepts

- The Harris response $R = \det(M) - k(\text{trace}(M))^2$ is computed per pixel; densely textured areas produce many pixels with high R , since local gradients vary sharply in many directions.
- Without additional filtering, every locally high response above the threshold is treated as a valid corner, regardless of its global significance or stability.
- Non-maximum suppression and adaptive thresholding are the main mechanisms used to reduce this redundancy while preserving the most significant keypoints.

4. Analysis

Excessive keypoints increase computational cost for matching (more descriptors to compute and compare) and reduce matching reliability, since many of the extra keypoints are low-quality, unstable, or correspond to noise-like local texture rather than meaningful structural features. This also increases the likelihood of false matches between images.

Method	Purpose
Non-Maximum Suppression	Keeps only the strongest corner in a local neighborhood
Adaptive / higher response threshold	Filters out weak, texture-induced corners
Shi-Tomasi 'good features to track'	Ranks corners by minimum eigenvalue, keeps top-N
Spatial distribution constraint (minDistance)	Enforces spacing between selected keypoints

5. Solution Design

- Apply non-maximum suppression in a local window to retain only the strongest corner response per neighborhood.
- Use `cv2.goodFeaturesToTrack` (Shi-Tomasi) with a `maxCorners` limit and `minDistance` parameter to cap and space out keypoints.
- Raise the corner response threshold so only strong, stable corners are retained.
- Rank keypoints by response strength and select only the top-N most significant ones for downstream matching.

6. Conclusion

By combining non-maximum suppression, adaptive thresholding, and a maximum keypoint limit with minimum spacing, the system can retain a manageable set of strong, stable corners from highly textured regions instead of an excessive, redundant set of low-quality key points.

Case Study 5: Feature Matching with Noise

1. Scenario

Feature matching between two images fails due to noise in one image.

2. Understanding of the Case

- Feature descriptors (e.g. SIFT, ORB) are computed from local gradient/intensity patterns around each keypoint.
- Noise perturbs pixel intensities, which distorts the local gradients used to build these descriptors, making them less representative of the true underlying structure.
- This causes descriptors from the noisy image to differ significantly from their true corresponding descriptors in the clean image, leading to missed or incorrect matches.

3. Application of Concepts

- Noise can also affect keypoint detection itself, generating spurious keypoints at noisy pixels or shifting the detected location of true keypoints.
- Descriptor distance metrics (e.g. Euclidean distance for SIFT, Hamming distance for ORB) become unreliable when descriptors are corrupted, since even matching features may exceed the distance threshold used to accept a match.
- Ratio-test based matching (Lowe's ratio test) can fail when noise makes the second-best match's distance similar to the best match's distance.

4. Analysis

Noise in one image reduces the repeatability of both keypoint detection and descriptor computation, so the descriptors extracted from the noisy image no longer correspond closely to descriptors from the same physical features in the clean image. This lowers the number of correct matches (true positives) and increases the number of incorrect matches (false positives), degrading overall matching accuracy.

Preprocessing Step	Benefit for Matching
Gaussian / Median Blur	Suppresses random noise while preserving edges
Bilateral Filter	Reduces noise while keeping strong structural edges sharp
Denoising (e.g. Non-Local Means)	Removes noise with minimal loss of fine detail
Stricter ratio test / RANSAC filtering	Removes remaining false matches after descriptor matching

5. Solution Design

- Apply a denoising filter (Gaussian blur, median filter, or bilateral filter) to the noisy image before keypoint detection.
- Use a more robust descriptor/detector combination if noise is expected to persist (e.g. SIFT tends to be more robust than simpler detectors).
- Apply Lowe's ratio test to discard ambiguous matches caused by noise-corrupted descriptors.
- Use RANSAC or a similar geometric consistency check after initial matching to remove outlier matches caused by noise.

6. Conclusion

Denoising the affected image before feature extraction, combined with robust matching strategies such as the ratio test and RANSAC-based outlier rejection, substantially improves feature matching accuracy in the presence of noise.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand